

Sana Firdous Abdul Kalam

Sharjah, UAE | +971 50 200 4587 | sanafirdous874@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Final-year Computer Science student at BITS Pilani Dubai (CGPA 9.41/10) specialising in AI/ML, computer vision, and full-stack development, graduating in 2026. Experienced in building end-to-end ML pipelines, computer vision systems, and LLM-powered applications across research and production settings. Hands-on with large-scale behavioural data and product analytics at LuLu Retail, serving over a million users. Seeking AI/ML, computer vision, or software engineering internship opportunities where I can apply and grow my skills.

EDUCATION

Birla Institute of Technology and Science (BITS Pilani), Dubai Campus <i>Bachelor of Engineering in Computer Science</i>	Sept 2022 – Sept 2026 CGPA: 9.41 / 10
Relevant Coursework: Large Language Models, Artificial Intelligence, Applied Statistical Methods, Data Mining, Foundations of Data Science, Probability & Statistics, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Optimization	
Our Own English High School, Sharjah <i>Higher Secondary Education - CBSE Class 12</i>	April 2008 – Aug 2022 94%

EXPERIENCE

AI Product & Analytics Intern <i>LuLu Retail</i>	January 2026 - Present Dubai, UAE
- Defined data requirements and feature specifications for an AI-driven personalisation and recommendation engine serving millions of users across mobile and desktop platforms, aligning with engineering on ML pipeline constraints and model trade-offs. - Identified cart-to-checkout as the critical 5% drop-off in the iOS/Android session-to-purchase conversion funnel through large-scale behavioural data analysis, directly shaping product decisions and prioritisation. - Conducted quantitative post-launch analysis for LuLu Money (live in-app remittance feature), surfacing usage anomalies and formulating a data-backed v2 proposal adopted by the product team. - Built KPI tracking and reporting frameworks covering order tracking, return/refund flows.	
Team Lead - Communication Systems <i>MAHASAT CubeSat Satellite Project, BITS Pilani Dubai</i>	Sept 2024 - Jan 2026
Leading CubeSat communication subsystem development - telemetry workflows, packet structures, and device-to-ground data transmission across hardware/software boundaries with rigorous signal processing and data integrity requirements.	
Professional Assistantship - C Programming Lab <i>BITS Pilani, Dubai Campus</i>	Sept 2025 – Jan 2026
Mentoring undergraduate students in debugging, logic-building, and structured problem-solving in C programming.	
Digitization Intern <i>Dubai Islamic Bank</i>	June 2024 – August 2024 Dubai, UAE
- Analysed digital channel adoption patterns to support data-driven feature prioritisation for the digital banking team. - Mapped customer onboarding journeys across ERP/CRM platforms, identifying process gaps through structured data analysis.	

PROJECTS

Trifecta Fusion Model – Terrain Segmentation <i>PyTorch, DeepLabV3+, MobileNetV3</i>	2025
- Proposed and implemented TFM, a novel multi-modal semantic segmentation architecture fusing RGB, depth, surface normals, and IMU via dynamic adaptive gating – submitted as a research report at BITS Pilani Dubai. - Achieved 52% relative improvement in cross-domain mIoU (0.395 → 0.600) under zero-shot transfer from Cityscapes to RELIS-3D off-road dataset, outperforming RGB-only and RGB+Depth baselines.	

- Designed a standardised multi-modal preprocessing pipeline across Cityscapes, KITTI Road, and RELIS-3D; conducted systematic ablation study isolating the contribution of each modality.

DoPilot - Multi-Agent AI Code Generator | *Python, LangGraph, LangChain, Gemini, Streamlit* 2025

- Built a LangGraph-orchestrated multi-agent pipeline (Planner → Architect → Coder → Security Agent → Verifier) that transforms natural language descriptions into production-ready codebases with security scanning and ZIP export.
- Implemented intelligent prompt enhancement via a guided 10-question Q&A workflow synthesising user input into a detailed technical specification before generation; live at dopilot.streamlit.app.

Intelligent Traffic Monitoring System | *Python, YOLOv8/v11, PyTorch, OpenCV* 2024

- Benchmarked five YOLO variants for real-time accident and vehicle detection; YOLOv8n achieved 91.7% mAP, 88.27% precision, 88.69% recall, outperforming the Roboflow baseline by +1.4% mAP.
- Built an automated SMTP email alert pipeline that triggers real-time notifications on accident or high-traffic detection; deployed with TensorRT optimisation for edge-device inference.

ASL Sign Language Detector | *Python, TensorFlow/Keras, MediaPipe, OpenCV, Streamlit* 2024

- Built a real-time ASL finger-spelling recognition system achieving 98%+ validation accuracy; custom LSTM trained on self-recorded dataset of 21 MediaPipe hand landmarks per frame (63 features).
- Implemented a 7-frame majority voting stabilisation mechanism reducing live inference flicker to under 50ms per frame; extended to support two simultaneous signers with independent sentence generation.

Deepfake Detection using Blockchain | *PyTorch, Xception, Grad-CAM, Solidity, Web3.py* 2024

- Built an end-to-end deepfake detection pipeline combining Xception-based CNN inference, Grad-CAM explainability heatmaps, and SHA-256 image hashing - providing both a prediction and a visual explanation for every result.
- Integrated an Ethereum smart contract (Solidity) to log every detection result immutably on-chain; designed four smart contract functions for model registration and tamper-proof detection retrieval.

Brain Tumor Detection - Segmentation & Classification | *PyTorch, U-Net, DenseNet, Streamlit* 2024

- Developed and benchmarked U-Net and DenseNet architectures for MRI-based brain tumor segmentation and classification; preprocessed and validated medical imaging datasets with consistent labelling and augmentation.
- Built an interactive Streamlit interface for clinical result visualisation, enabling side-by-side comparison of segmentation outputs and model performance metrics.

TECHNICAL SKILLS

Languages: Python, SQL, C, C++, JavaScript, Java, HTML/CSS

AI / ML / CV: PyTorch, TensorFlow, Keras, scikit-learn, YOLOv8/v11, DeepLabV3+, MobileNetV3, LSTMs, CNNs, Transformers, RAG, FAISS, Grad-CAM, MediaPipe, OpenCV

LLM & Agents: Google Gemini, Groq API, LLaMA, LangChain, LangGraph, MCP Servers

Backend & APIs: FastAPI, REST APIs, SQLAlchemy, PostgreSQL, SQLite

Frontend: Next.js 14, React, TypeScript, Tailwind CSS, Streamlit

Data: Pandas, NumPy, ETL pipelines, Power BI, Tableau

Cloud & Tools: Azure, Google Cloud, Git/GitHub, Docker (basic), Postman, Jupyter, VS Code

Blockchain: Solidity, Web3.py, Ethereum smart contracts

CERTIFICATIONS

AI Engineering Career Readiness Program (8 weeks) – Tecoholic × BITS Pilani | *Consistent Performer* Oct–Dec 2025

AI Agents and Agentic AI with Python – Vanderbilt University

Google Cloud Generative AI Learning Path – Google Cloud

Azure Cloud Computing Fundamentals – Microsoft Learn

Deloitte Audit and Assurance Virtual Internship - Forage

Power BI Fundamentals – Simplilearn

VOLUNTEERING

- **Host - ICAIN 2025 International Conference on Artificial Intelligence:** Moderated keynote and panel sessions, coordinated logistics with speakers, and ensured seamless event execution.
- **Fundraising Intern - Rampur Education and Welfare Trust (REWT, NGO):** Led digital outreach campaigns and maintained donor relations, boosting engagement and fundraising impact.